

PATENSIE, South Africa

WMO: 688350

Lat: 33.7653S

Lon: 24.8233E

Elev: 97

StdP: 100.17

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.6	5.9	-2.6	3.1	19.1	-0.7	3.6	17.7	8.2	20.8	7.2	20.5	0.9	340	0.300

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.7	33.1	20.9	31.1	20.6	29.5	20.1	23.0	28.6	22.4	27.8	21.7	26.9	3.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.5	16.4	24.6	20.8	15.6	24.3	20.0	14.9	23.7	68.7	28.3	66.2	28.3	63.6	26.9	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.6	5.4	4.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	22.6	23.1	21.7	19.0	17.3	14.6	14.5	14.6	16.1	18.1	19.7	22.0
	DBStd	4.13	2.49	2.38	2.72	3.09	2.69	2.88	3.12	2.59	2.59	2.86	2.78	2.68
	HDD10.0	3	0	0	0	0	0	1	2	1	0	0	0	0
	HDD18.3	583	1	1	3	27	54	118	127	118	76	39	17	2
	CDD10.0	3134	389	366	363	269	226	139	141	144	183	250	291	372
	CDD18.3	674	132	133	109	47	21	7	8	4	9	31	57	116
	CDH23.3	4183	791	755	593	267	153	77	88	51	79	222	336	772
	CDH26.7	1264	247	242	174	87	35	13	14	8	27	74	85	258
Wind	WSAvg	1.5	1.5	1.4	1.3	1.2	1.3	1.6	1.7	1.5	1.5	1.5	1.7	1.6
Precipitation	PrecAvg	577	43	42	62	50	34	35	48	57	44	52	53	45
	PrecMax	907	99	113	167	183	122	120	120	200	134	142	131	192
	PrecMin	430	10	16	13	15	6	7	7	10	8	17	10	8
	PrecStd	124	26	27	40	40	29	27	38	51	35	32	36	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.5	34.5	33.7	33.6	30.0	28.5	28.4	27.9	31.7	33.8	32.1	35.0
		MCWB	21.7	22.1	21.3	19.3	16.2	14.4	14.3	14.7	16.6	19.3	20.0	21.5
	2%	DB	31.5	31.5	30.7	28.8	27.0	25.6	25.6	24.5	25.0	28.2	29.2	31.5
		MCWB	21.9	21.8	20.7	18.4	16.1	13.2	13.4	14.1	15.7	18.7	19.6	21.1
	5%	DB	29.5	29.5	28.5	26.1	24.6	23.1	23.1	21.9	22.8	25.5	27.1	29.2
		MCWB	20.9	21.2	20.2	18.1	15.8	12.8	13.2	14.0	15.4	17.5	18.6	20.5
	10%	DB	27.8	27.7	26.7	23.9	22.3	20.6	20.5	20.0	20.9	23.2	25.1	27.3
		MCWB	20.1	20.5	19.7	17.2	15.5	12.3	12.7	13.2	14.8	16.6	18.0	19.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.9	23.7	23.2	21.1	19.1	16.5	16.2	17.8	18.7	21.3	21.7	23.4
		MCDB	30.0	30.1	27.8	27.7	23.7	22.3	21.8	23.6	25.0	29.4	29.1	30.1
	2%	WB	22.9	22.8	22.4	19.8	17.7	15.2	15.2	16.2	17.3	19.5	20.5	22.4
		MCDB	28.2	28.6	27.2	26.7	22.2	20.4	21.2	20.2	22.8	25.5	26.7	28.5
	5%	WB	22.0	22.1	21.5	19.0	16.9	14.2	14.5	15.2	16.3	18.4	19.6	21.5
		MCDB	27.4	27.3	26.3	24.5	21.8	19.9	20.3	18.7	21.3	23.6	24.8	27.3
	10%	WB	21.1	21.4	20.6	18.1	16.1	13.4	13.7	14.1	15.2	17.4	18.8	20.7
		MCDB	26.3	26.1	25.2	22.7	20.8	18.8	19.2	18.1	20.0	22.1	23.7	25.7
Mean Daily Temperature Range	5% DB	MDBR	9.8	9.7	9.9	10.3	11.4	11.9	11.5	11.5	10.5	10.0	9.8	10.3
		MCDBR	12.9	13.0	13.2	14.0	14.6	15.3	15.4	14.7	13.8	14.7	13.8	14.3
		MCWBR	5.0	4.9	4.9	5.6	5.8	6.2	5.9	6.9	6.3	6.9	5.8	5.5
	5% WB	MCDBR	11.1	10.9	11.2	12.2	11.7	12.1	12.2	11.1	12.4	12.6	11.8	12.5
		MCWBR	4.7	4.3	4.5	5.2	5.6	6.1	5.9	6.3	6.2	6.4	5.4	4.8
Clear-Sky Solar Irradiance	taub	0.378	0.382	0.366	0.342	0.326	0.313	0.315	0.350	0.370	0.386	0.368	0.374	
	taud	2.478	2.473	2.498	2.543	2.559	2.568	2.556	2.449	2.387	2.379	2.463	2.476	
	Ebn at Noon	960	935	912	881	843	829	846	856	893	920	963	967	
	Edh at Noon	117	113	103	88	76	71	75	95	113	123	118	118	
All-Sky Solar Radiation	RadAvg	7.12	6.36	5.30	4.03	3.09	2.70	2.92	3.73	4.92	5.92	6.86	7.39	
	RadStd	0.33	0.32	0.36	0.20	0.14	0.15	0.18	0.23	0.29	0.41	0.46	0.49	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (8 neighbors)		+0.38	+0.69	+0.84	N/A	N/A	N/A	N/A	-85	+142	+45	

Nomenclature: See separate page